

Project Report

Project Title: Personal Expense Tracker

Domain: Python Programming / Financial Management System.

Personal Expense Tracker is a simple application designed to help users monitor and manage their daily expenses. It allows users to input multiple expenses, calculates the total spending, and categorizes it into Low, Moderate, or High based on predefined thresholds.

2. Introduction

Managing personal finances is an essential life skill. Many individuals struggle to keep track of their daily expenses, leading to poor financial planning. The Personal Expense Tracker provides a simple solution by allowing users to record expenses and analyze their spending patterns.

The goal of this project is to build a basic system that processes expense data, calculates totals, and provides meaningful insights using simple logic and conditions

3. Objectives of the Project

- Track daily expenses efficiently
- Calculate total spending automatically
- Classify expenses into categories (Low, Moderate, High)
- Provide quick insights into spending habits
- Develop a simple and user-friendly program
- Strengthen programming logic and problem-solving skills

4. System Overview

The Personal Expense Tracker is a console-based application developed using Python.

- Users input the number of expenses
- Enter individual expense amounts
- The system calculates total expenses
- Based on thresholds, it classifies spending level
- Displays total expense and category clearly

5. Modules and Concepts Applied

5.1 Modules

- **Input Module** – Accepts expense values from the user
- **Calculation Module** – Computes total expense
- **Classification Module** – Categorizes expenses
- **Output Module** – Displays results to the user

5.2 Concepts Applied

- Python programming fundamentals
- Functions for modular design
- Loops for handling multiple inputs
- Conditional statements for classification

- Basic arithmetic operations
- User input handling

6. Features Implemented

- Accept multiple expense entries
- Calculate total spending
- Categorize expenses (Low, Moderate, High)
- Simple and clean user interface (console-based)
- Fast and efficient execution
- Beginner-friendly design

7. System Architecture

Source Code Structure:

Personal-Expense-Tracker

├— expense_tracker.py # Main program logic

├— README.md # Project description

- **expense_tracker.py:** Handles input, processing, and output
- **README.md:** Provides project overview and instructions

8. Implementation Details (File-wise Description)

expense_tracker.py

Handles:

- Taking number of expenses as input
- Looping through each expense entry
- Calculating total expense
- Classifying expense category using conditions
- Displaying final results

9. Challenges Faced and Solutions

- Handling multiple inputs – Solved using loops
- Accurate total calculation – Used proper data types (float)
- Classification logic – Implemented clear conditional structure
- User input errors – Assumed valid input for simplicity

10. Testing

Tested features include:

- Correct calculation of total expenses
- Proper classification of spending levels
- Handling multiple expense inputs
- Output display accuracy

All tests produced correct and expected results.

11. Future Enhancements

- Add expense categories (food, travel, etc.)
- Save data using files or databases
- Graphical representation of expenses
- Monthly and yearly reports
- Mobile or web-based interface
- Budget limit alerts

Results

```
File Edit Format Run Options Window Help
# Personal Expense Tracker

def check_expense_category(total_expense):
    if total_expense <= 2000:
        return "Low"
    elif total_expense <= 5000:
        return "Moderate"
    else:
        return "High"

# Taking number of expenses
n = int(input("Enter number of expenses: "))

total = 0

# Taking expense inputs
for i in range(n):
    expense = float(input(f"Enter expense {i+1}: "))
    total += expense

# Checking category
category = check_expense_category(total)

# Display result
print("Total Expense:", total)
print("Expense Category:", category)
```

```
File Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

- RESTART: C:/Users/Jalaka Uma/OneDrive/Desktop/Personal Expense Tracker Task 1/personal_expense_tracker.py
Enter number of expenses: 3
Enter expense 1: 200
Enter expense 2: 900
Enter expense 3: 900
Total Expense: 2000.0
Expense Category: Low
```

12. Conclusion

The Personal Expense Tracker provides a simple and effective way to monitor daily spending. It demonstrates how basic programming concepts such as loops, functions, and conditional statements can be used to solve real-life financial problems. The system processes user input efficiently and provides meaningful output through classification of expenses.

Although simple, the project reflects practical applications in personal finance management and can be extended with advanced features for better usability and scalability. Overall, it enhances logical thinking and showcases the use of programming in everyday life

13. References

- Python documentation
- Basic programming tutorials
- Financial management concepts
- Project-based learning resources